

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	A. Mohith krishna	Reg. No.:	192425199
Section / Batch:	2024	Date:	30/08/2026

Code of Conduct

I A. Mohith Krishna (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A. Mohith

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

- 1. Maintain entity coherence using assignment, mistakes, and you.**
- 2. Use a Cause–Effect or Elaboration discourse relation.**
- 3. Include at least two keywords from:**
 - **review**
 - **improve**
 - **confident**

- feedback
- 4. Response length must be 2–3 sentences.
- 5. Maintain a positive and polite tone.
- 6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a

river Constraints:

- 1. Resolve the correct word sense using contextual information.
- 2. Identify important entities and actions.
- 3. Represent the meaning using predicate logic.
- 4. Preserve semantic relationships.
- 5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.

c) Write a predicate logic representation using suitable predicates

such as: sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

1. Reference Resolution and Discourse Coherence

Given Paragraph

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Referring Expressions and Possible Antecedents

The referring expressions are:

- He → Ravi / Arun
- him → Ravi / Arun
- it → the book
- they → Ravi and Arun
- the book → the book introduced earlier
- they in the final sentence → Ravi and Arun

b) Applying Reference Resolution Constraints

The following constraints are used:

1. Gender and number agreement – The pronoun must agree with the antecedent.
2. Recency – A recently mentioned entity is generally preferred.
3. Grammatical role – Subject/object roles provide clues.
4. Semantic compatibility – The reference must make logical sense.
5. Discourse coherence – The interpretation should make the whole paragraph meaningful.

c) Reference Resolution Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Male, singular, and grammatically suitable
He	Arun	Valid	Male and singular; also recently mentioned
him	Ravi	Valid	Male, singular, and can be helped

him	Arun	Invalid	Arun is the grammatical subject of “helped,” so referring to himself is unlikely
it	book	Valid	Singular and semantically compatible with “find”
it	Artificial Intelligence	Invalid	“Find Artificial Intelligence” is not the intended meaning
they	Ravi + Arun	Valid	Plural pronoun referring to both people
the book	Previously mentioned book	Valid	Maintains discourse continuity
they	Ravi + Arun	Valid	Both participants are involved in the discussion

Resolving “He”

The sentence:

“He was looking for a book on Artificial Intelligence.”

Both Ravi and Arun are grammatically possible because both are male and singular.

However, Ravi is the preferred antecedent because he is the subject of the previous sentence:

Ravi met Arun at the library.

The interpretation is therefore:

Ravi was looking for a book on Artificial Intelligence.

Then:

Arun helped him find it.

Here, him = Ravi and it = the book.

d) Final Coreference Chains

Person Chain 1

Ravi → He → him

Person Chain 2

Arun → Arun

Person Chain 3

Ravi + Arun → they → they

Object Chain

a book → it → the book

e) Paragraph with Pronouns Replaced

Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

f) Ambiguity if “He was looking for a book” Appeared Before Introducing Arun

If the sentence appeared before Arun was introduced:

“Ravi met someone at the library. He was looking for a book.”

the pronoun “He” would have fewer possible antecedents because Arun has not yet been introduced.

However, if the sentence were:

“He was looking for a book. Ravi met Arun at the library.”

then He has no clearly introduced antecedent. The reader cannot determine whether He refers to Ravi, Arun, or another person.

Therefore, introducing the entities before using pronouns improves reference resolution and discourse coherence.

2. Dialog Act Recognition and Response Generation

Given Conversation

Student:

“I submitted my assignment, but I am worried that I may have made many mistakes.”

a) Required Dialog Acts

The required dialog acts are:

1. Reassure

The response should reduce the student's worry and provide confidence.

2. Advise

The response should provide a useful action, such as reviewing the assignment or using feedback.

Therefore:

Dialog Act = Reassure + Advise

b) Important Entities

The important entities that must remain consistent are:

Entity

Role

assignment Work submitted by the student

mistakes Possible errors in the assignment

you The student receiving the response

The response should continue discussing the same assignment and mistakes rather than introducing unrelated topics.

c) Three Possible Responses

Response 1 – Balanced

It is understandable to feel worried after submitting an assignment, but you can be confident that reviewing your mistakes can help you improve. Use the feedback you receive to identify areas for improvement and strengthen your future assignments.

Response 2 – Encouraging

Don't worry too much about the assignment; mistakes are a natural part of learning. Review the feedback carefully so you can improve your understanding and feel more confident in your next assignment.

Response 3 – Direct

You have already completed and submitted the assignment, so try to stay confident rather than focusing only on possible mistakes. Review the assignment when possible and use feedback to improve your future work.

d) Comparison of Responses

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
1	High	High	High	High

2	High	High	High	High
3	High	High	High	High

Detailed Comparison

Response 1:

Maintains strong entity coherence and clearly connects mistakes → review → improvement. It satisfies the reassurance and advice requirements.

Response 2:

Has a friendly and encouraging tone. It uses review, feedback, improve, and confident, satisfying the keyword requirement.

Response 3:

Provides practical advice and directly addresses the student's concern. It also avoids contradictory advice.

e) Best Response

Response 1 is the best choice because it provides a natural combination of reassurance and practical advice. It maintains the entities assignment, mistakes, and you, uses an elaboration/cause-effect relationship, and maintains a positive tone.

f) What Happens if Entity Coherence Is Not Maintained?

If entity coherence is not maintained, the response may become confusing or irrelevant.

For example:

“Your project looks good. Review your exam results to improve your performance.”

The conversation was about an assignment, but the response suddenly discusses exam results. This causes:

- Loss of context
- Reduced relevance
- Confusion about the topic
- Poor conversational quality

Therefore, maintaining entity coherence is essential for generating meaningful responses.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence

“Priya sat on the bank and watched the boats moving across the water.”

a) Possible Meanings of “Bank”

The word bank has two possible meanings:

Sense	Meaning
Financial bank	An institution that manages money and provides financial services
River bank	The land alongside a river or body of water

b) Correct Sense

The correct meaning is:

Bank = the side of a river

Reason

The surrounding words provide strong contextual evidence:

- sat on the bank

- boats
- moving
- water

Boats move across water, and a person can sit on the bank of a river and watch boats.

The financial interpretation is semantically incompatible with the context because:

“Priya sat on the financial institution and watched boats...”

does not make sense in the intended interpretation.

Therefore:

Selected sense = River bank

c) Predicate Logic Representation

Let:

- Priya = p
- Bank = b
- Boat = z
- Water = w

A suitable predicate representation is:

sit(p, b)
 bank(b)
 beside(p, b)
 watch(p, z)
 boat(z)
 move(z)
 water(w)
 across(z, w)

A more complete logical representation can be written as:

$$\begin{aligned} &\exists p \exists b \exists z \exists w [\\ &\quad \text{person}(p) \wedge \\ &\quad \text{bank}(b) \wedge \\ &\quad \text{water}(w) \wedge \\ &\quad \text{boat}(z) \wedge \\ &\quad \text{sit}(p,b) \wedge \\ &\quad \text{beside}(p,b) \wedge \\ &\quad \text{watch}(p,z) \wedge \\ &\quad \text{move}(z) \wedge \\ &\quad \text{across}(z,w) \\ &] \end{aligned}$$

Meaning

This represents that:

- Priya is a person.
- There is a river bank.
- Priya sits beside/on the bank.
- There is a boat.
- Priya watches the boat.
- The boat is moving across the water.

d) Unambiguous English Paraphrase

Priya sat on the side of the river and watched the boats moving across the water.

This paraphrase removes the ambiguity of bank by explicitly saying “the side of the river.”

e) How Word Sense Disambiguation Improves Machine Translation

Word Sense Disambiguation (WSD) determines the correct meaning of a word based on its context before translation.

For example, the word “bank” could be translated differently depending on its meaning.

Without WSD

The translation system may incorrectly interpret:

bank → financial institution

This could produce an incorrect translation.

With WSD

The system recognizes the context:

boats + water + sat on

and selects:

bank → side of a river

Therefore, WSD helps machine translation by:

- Selecting the correct word meaning.
- Reducing translation errors.
- Preserving semantic relationships.

- Improving contextual accuracy.
- Producing more natural translations.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding ; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints

Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application ; requires guidance	Unable to apply relevant concepts
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Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided
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Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	4	4	3	3	3		83
2	3	3	4	4	3		
3	3	3	3	4	4		

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____